

Design and Implementation of Power Distribution System for a New Town Along the Lahore-Islamabad Motorway



(Department of Electrical and Computer Engineering)

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Subject: Power Distribution and Utilization

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Submission Date: *25th December , 2024*

Abstract

This report presents a comprehensive proposal for the design and implementation of a power distribution system for a newly planned town located along the Lahore-Islamabad motorway. The town, spanning an area of 220 square kilometers and anticipated to accommodate 10,000 families along with numerous public and industrial facilities, requires a robust power supply infrastructure.

Given its location 16 kilometers away from the nearest 500 kV national grid, a new substation and distribution network are essential to meet both current and future demand.

The proposal includes a detailed load breakdown, power consumption calculations, and the design of the substation and distribution network. The report also incorporates protection schemes, power factor improvement strategies, and ETAP simulations for system performance evaluation. The final solution ensures a reliable, efficient, and scalable power supply to meet the growing needs of the town, ensuring its future growth and sustainability.

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2. Executive Summary

This report outlines the power distribution design proposal for a newly planned town located along the Lahore-Islamabad motorway. With a projected population of 10,000 families, the town includes various sectors with residential, healthcare, educational, industrial, and commercial facilities. Due to its distance from the nearest 500 kV national grid (16 km away), an efficient and reliable power supply solution is essential.

The proposal includes:

- Detailed load analysis, power consumption calculations, and sector-specific demand.
- Design of the substation and distribution network.
- Protection schemes for both substation and distribution lines.
- Power factor improvement strategies to enhance system efficiency.
- Simulation and testing using ETAP software to evaluate system performance under different conditions.

The final design aims to provide a reliable, cost-effective, and sustainable power distribution solution to meet the needs of the town.

3. Introduction

The proposed town, located along the Lahore-Islamabad motorway, represents a major development project with significant residential, commercial, and industrial infrastructure. Given its remote location (16 km away from the nearest national grid), it is essential to develop a robust power distribution system to meet both immediate and future needs.

This report presents a detailed proposal for the power distribution system, covering:

- Site analysis and load breakdown.
- Substation design, including incoming and outgoing lines, transformer configurations, and substation components.

- Distribution system design, including feeder layout, conductor sizing, and voltage sag calculations.
- Protection schemes for the substation and distribution network to ensure safe and reliable operation.
- Power factor improvement through capacitor bank sizing and placement.
- Simulation and testing of the proposed system using ETAP software to validate the design.

The overall goal is to provide a sustainable, scalable, and efficient power distribution system for the proposed town.

4. Site Overview and Load Breakdown

Site Overview

The proposed town covers 220 square kilometers, divided into six sectors. These sectors will house a mix of residential, commercial, healthcare, educational, and industrial facilities, each with unique power demands.

Sector Load Breakdown

The town is divided into six sectors (A to F), each with specific load characteristics. The breakdown for each sector is as follows:

Sectors A, B, C, D (Each sector):

- **Residential Load:** 2000 houses
- **Healthcare:** 1 hospital
- **Religious:** 25 mosques
- **Commercial:** 3 markets
- **Industrial:** 1 small factory
- **Education:** 10 schools, 5 colleges, 2 universities

Sectors E, F (Each sector):

- **Residential Load:** 1000 houses
- **Healthcare:** 1 hospital
- **Religious:** 25 mosques
- **Commercial:** 4 markets, 1 mega project, 1 theme park
- **Industrial:** 1 small factory
- **Water Treatment:** 1 water treatment plant
- **Education:** 10 schools, 5 colleges, 2 universities

Sector A has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector A.

Sector B has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector B.

Sector C has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector C.

Sector D has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector D.

Sector E has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector E.

Sector F has Two outgoing 11 kV line (24 feeders) allocate to feed Power to Sector F.

Loads equally distribute among all sectors :

Table :

	Sector A	Sector B	Sector C	Sector D	Sector E	Sector F
Houses	2000	2000	2000	2000	1000	1000
Hospitals	1	1	1	1	2	2
Masjid	25	25	25	25	25	25
Markets	3	3	3	3	4	4
Factories	1	1	0	1	1	1
Theme Parks	0	0	0	0	1	1
Mega Projects	0	0	0	0	1	1
Schools	10	10	10	10	10	10
Colleges	5	5	5	5	5	5
Universities	2	2	2	2	2	2
Water Treatment Plant	0	0	0	0	1	1

This breakdown helps estimate the total power consumption, which is used to determine transformer ratings, feeder capacities, and other system components.

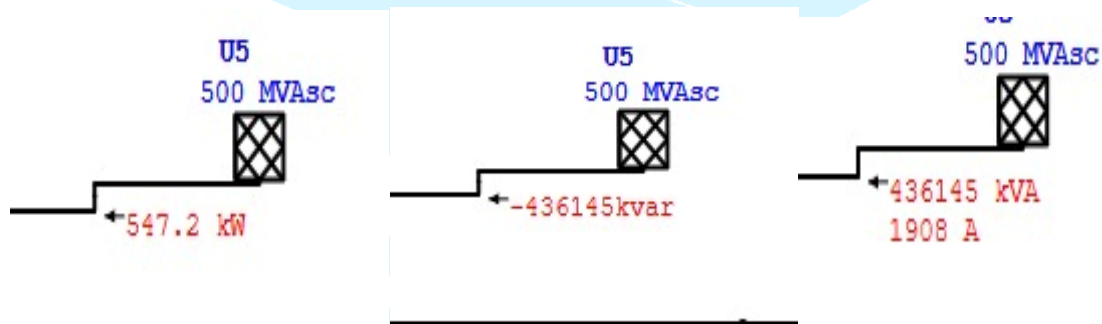
5. Power Consumption Calculations

Total Load per Sector

The total power consumption is calculated for each sector based on the type and number of facilities. The residential, healthcare, commercial, and educational components contribute significantly to the load.

For example:

- **Sectors A, B, C, D:** Each sector has a total power demand of **3725 kW**.
- **Sectors E, F:** Each sector has a total power demand of **4750 kW**.



These upper power demand ensure that the power system is designed to meet both the present and future demands of the town while accommodating potential growth in residential and industrial sectors.

6. Substation Design

6.1 Incoming and Outgoing Lines

The substation will have **two incoming 132 kV lines** from the nearby 500 kV grid station, which will step down the voltage from 132 kV to 11kV to provide the necessary power for the town's distribution network.

From the substation, there will be **two outgoing 132 kV lines** that connect to another grid station, ensuring system reliability and redundancy in the power supply network.

Additionally, the substation will distribute power through **12 outgoing 11 kV lines**. Each **11 kV line** will feed **12 feeders** in total, which will be responsible for delivering power to various sectors in the town. The layout will be as follows:

- **Each 11 kV line will feed 12 feeders**, making a total of **12 feeders** from each of the 11 kV lines.
- The feeders will be distributed across the six sectors, ensuring each sector receives its required power from the substation.

The power distribution network will be carefully designed to balance the load across all feeders and sectors, minimizing losses and maintaining voltage levels within acceptable limits. This system layout ensures the town will have sufficient power supply and reliability for its residential, industrial, commercial, and public services.

6.2 Transformer Configuration

The substation will use step-down transformers capable of handling approximately **31.5 MW to 40 MW per sector** of load, accounting for both diversity and safety margins.

6.3 Substation Components

The substation will feature:

- **Busbar Configuration:** Ensures flexibility and reliability.
- **Switchgear and Circuit Breakers:** For fault detection and system isolation.
- **Monitoring and Control Systems:** SCADA for real-time monitoring and management.

7. Distribution System Design

7.1 Feeder Design

A total of 12 outgoing feeders will supply power to the town, each feeder connected to a sector's 11 kV network. The feeders are designed for minimal losses, with appropriate conductor sizes based on current density and voltage drop calculations.

7.2 Voltage Sag Calculations

Voltage sag calculations along the longest feeder ensure the voltage drop does not exceed 5%. These calculations consider load growth, impedance, and feeder length.

The input to the street transformer is exactly 11 kV . So the output of the power transformer placed in Distribution Substation is 11.7 kV so the total length of each feeder is maximum 22 Km .

8. Protection Scheme

8.1 Substation Protection

The substation will be equipped with:

- **Differential Protection** for transformers.
- **Overcurrent Protection** to detect and isolate faults.
- **Busbar Protection** for quick fault isolation within the busbar.

8.2 Distribution Network Protection

Each distribution feeder will have:

- **Overcurrent and Earth Fault Protection.**
- **Fuse Coordination** for selective fault isolation.
- **Automatic Reclosers** to minimize system downtime.

9. Power Factor Improvement

9.1 Capacitor Bank Sizing

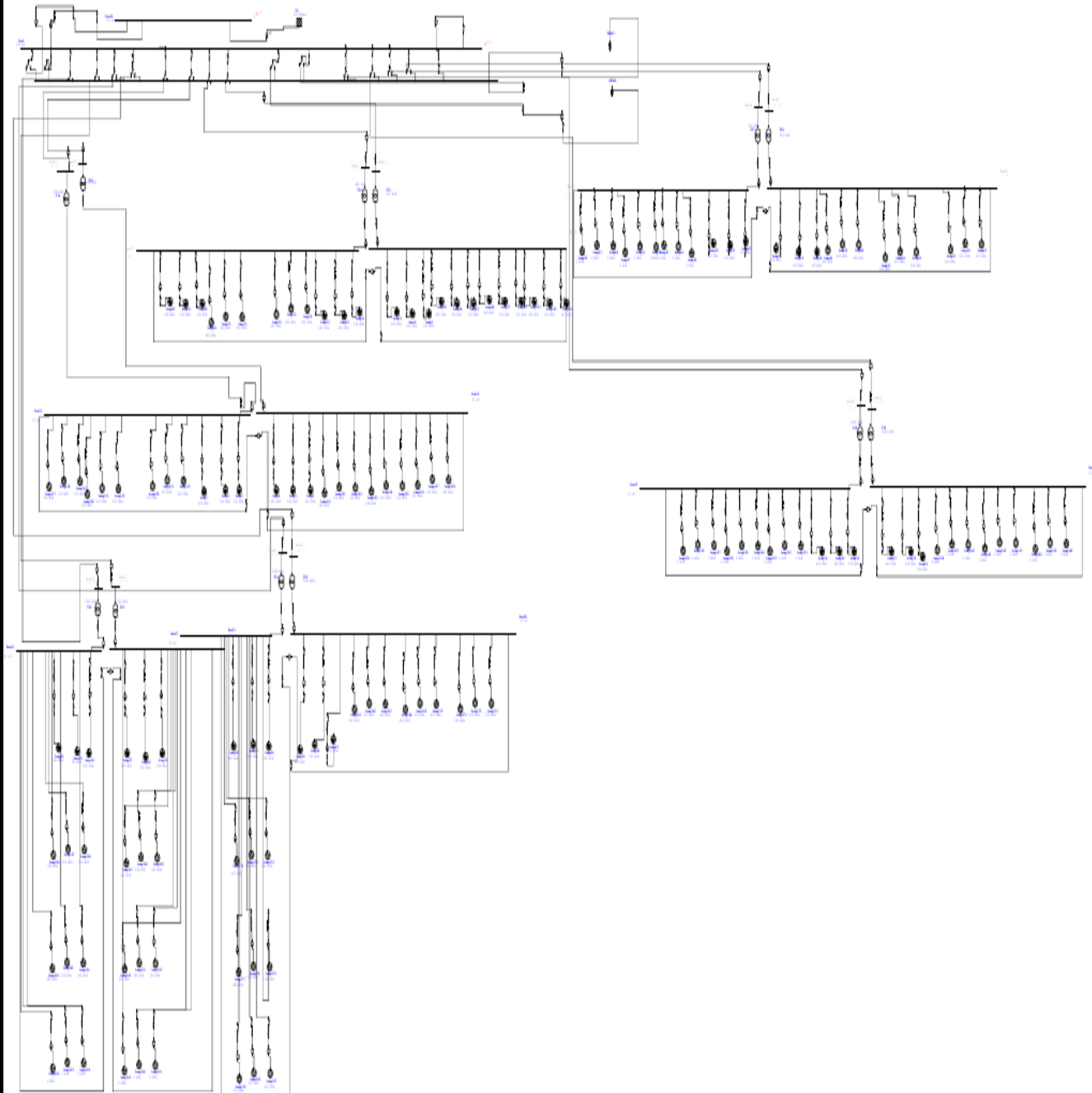
Capacitor banks will be sized to provide reactive power compensation, improving the power factor of the network and reducing losses. The sizing is based on the reactive power demand and system characteristics.

Before Power factor correction the power factor is 0.7 lagging so I also find the reactive and apparent power of 0.7 power factor lagging and same process used to find after power factor correction the power factor is 0.9 lagging and find the reactive or apparent power of power factor 0.9 lagging. So that subtraction of reactive power of 0.7 power factor and reactive power of 0.9 power factor. This result is that whose reactive power produced by capacitor. So that decrease the reactive power of system as a result losses of the system is reduced so this system is more efficient. I find capacitance value of capacitor through (Q_c) reactive power produced by capacitor. So I find the capacitance of the capacitor which I placed by capacitor bank.

10. Simulation and Testing

10.1 ETAP Simulation

The power system model was created in ETAP to perform load flow analysis, fault analysis, and voltage stability studies. This simulation ensures the system can handle varying loads and fault conditions.

ETAP Simulation:**One-Line Diagram - OLV1 (Load Flow Analysis)**

ALERT Summary Report :

Project:	ETAP	Page:	1
Location:	19.0.1C	Date:	12-22-2024
Contract:		SN:	
Engineers :Muhammad shahzaib /Hamza Nasir /Muhamad Ali	Study Case: Load Flow	Revision:	Base
Filename: substation design		Config:	Normal

Alert Summary Report

	% Alert Settings	
	Critical	Marginal
<u>Loading</u>		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
<u>Bus Voltage</u>		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
<u>Generator Excitation</u>		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
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BUSBAR Loading Summary Report :

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic		MVA	% PF	Amp	Percent Loading
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar				
Bus1	132.000													
Bus48	132.000													
Bus52	132.000													

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).

Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Bus Input Data

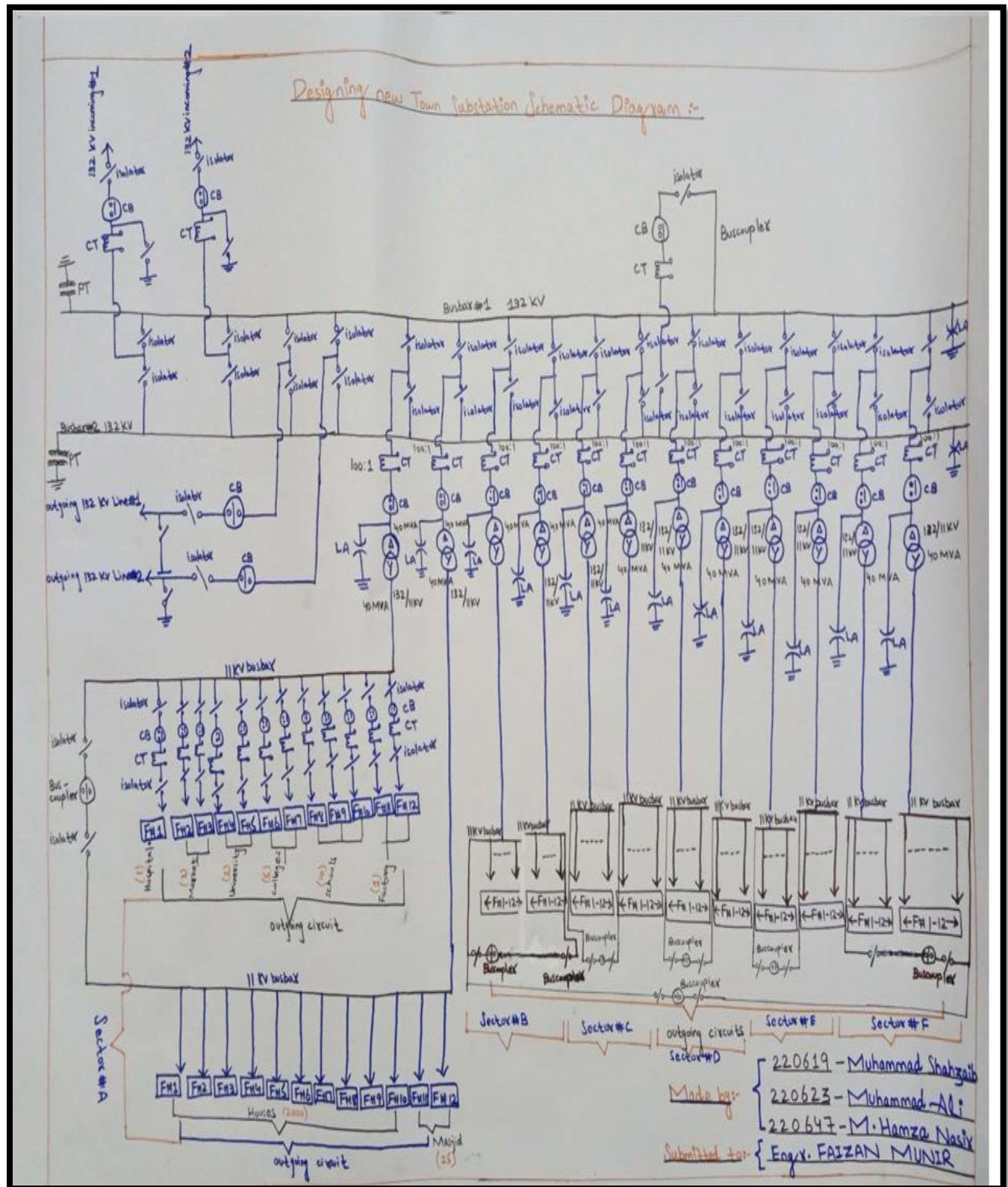
Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-bus	% Mag.	Angle	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	132.000	1	100.0	30.0								
Bus48	132.000	1	100.0	0.0								
Bus52	132.000	1	100.0	0.0								

Total Number of Buses: 3

0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.000

Generation Bus			Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-bus	% Mag.	Angle	MW	Mvar	% PF	Max. Min.
Bus52	132.000	Swing	1	100.0	0.0				
						0.000	0.000		

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10.2 Protection Scheme Testing

The protection scheme was tested using different fault scenarios (bus-to-bus, line-to-line) to confirm the system's fault detection and isolation capabilities. This testing ensures a high level of reliability and safety.

11. Conclusion

In conclusion, the proposed power distribution system for the newly planned town along the Lahore-Islamabad motorway represents a comprehensive and scalable solution to meet the diverse needs of the community. The system's design—covering everything from load calculations and transformer sizing to distribution and protection schemes—ensures that the infrastructure is capable of handling current demand and future growth.

The inclusion of advanced simulation techniques (using ETAP software) and the testing of protection schemes guarantees that the system will function efficiently under normal and fault conditions, minimizing downtime and enhancing safety. The design also integrates power factor improvement strategies, ensuring that the system operates efficiently while maintaining grid stability.

Overall, this proposal provides a robust, cost-effective, and reliable solution for the power supply needs of the town. By adhering to industry standards and incorporating modern technology, the project will ensure long-term sustainability and support the town's development for years to come.

12. References

1. IEEE Standard for Power System Protection.
2. ETAP Software Documentation.
3. "Power System Protection and Switchgear" by Badri Ram, D.N. Vishwakarma.
4. Relevant National Grid and Distribution Network Guidelines from the Pakistan National Power Control Center.